

Arduino Code for the Environmental Monitoring System

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#include <Wire.h>
#include <SPI.h>
#include <SD.h>
#include <Adafruit_SCD30.h>
#include <Adafruit_SHT4x.h>

Adafruit_SCD30 scd30;
Adafruit_SHT4x sht4;

const int chipSelect = 10;
unsigned long startTime;

void setup() {

  Serial.begin(9600);

  scd30.begin();
  sht4.begin();

  SD.begin(chipSelect);

  pinMode(LED_BUILTIN, OUTPUT);

  File dataFile = SD.open("data.csv", FILE_WRITE);

  if (dataFile) {
    dataFile.println("Time,CO2,SCD30_Temp,SCD30_RH,SHT_Temp,SHT_RH");
    dataFile.close();
  }

  startTime = millis();
}

void loop() {

  if (millis() - startTime < 100000) {

    // LED ON
    digitalWrite(LED_BUILTIN, HIGH);
    delay(5000);

    // Read SCD30
    scd30.read();

    // Read SHT40
    sensors_event_t humidity, temp;
    sht4.getEvent(&humidity, &temp);

    // Print to Serial Monitor
    Serial.print("CO2: ");
```

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Serial.print(scd30.CO2);

Serial.print(" | SCD30 T: ");
Serial.print(scd30.temperature);

Serial.print(" | SCD30 RH: ");
Serial.print(scd30.relative_humidity);

Serial.print(" | SHT T: ");
Serial.print(temp.temperature);

Serial.print(" | SHT RH: ");
Serial.println(humidity.relative_humidity);

// Save to SD card
File dataFile = SD.open("data.csv", FILE_WRITE);

if (dataFile) {

    dataFile.print(millis());
    dataFile.print(",");

    dataFile.print(scd30.CO2);
    dataFile.print(",");

    dataFile.print(scd30.temperature);
    dataFile.print(",");

    dataFile.print(scd30.relative_humidity);
    dataFile.print(",");

    dataFile.print(temp.temperature);
    dataFile.print(",");

    dataFile.println(humidity.relative_humidity);

    dataFile.close();
}

// LED OFF
digitalWrite(LED_BUILTIN, LOW);
delay(5000);

}
else {

    Serial.println("Program stopped.");

    while (1);
}
}

```

